

Gridtopia Gerrymandering

Gridtopia and Gerrymandering

On the following page, you will see voters (hearts and clubs) in a 7×7 grid along with a 7×7 district plan with 7 districts. (Today's lesson brought to you by the [number seven](#) ([sound on!](#)).)

Questions

Define $p(D)$ to be the perimeter of a district D and $a(D)$ to be the area of a district D .

1. Compute the Polsby-Popper score for each district,

$$pp(D) = \frac{4\pi a(D)}{p(D)^2}.$$

2. Compute the Schwarzberger score for each district,

$$s(D) = \frac{c(D)}{p(D)},$$

where $c(D)$ is the circumference of a circle of area $a(D)$.

3. Compute the Convex Hull score for each district,

$$ch(D) = \frac{a(D)}{a(MP(D))},$$

where $MP(D)$ is the smallest polygon containing the district D .

4. Compute the Reock score for each district,

$$r(D) = \frac{a(D)}{a(MC(D))},$$

where $MC(D)$ is the smallest circle containing D ; the previous exercise might help you figure out the diameter of this circle.

5. Compute each district's x - and y -symmetry scores,

$$xs(D) = \frac{a(f(D) \cap D)}{a(D)},$$

(respectively $ys(D)$) where $a(f(D) \cap D)$ is the area in common between D and a reflection $f(D)$ across the horizontal axis (respectively, vertical axis).

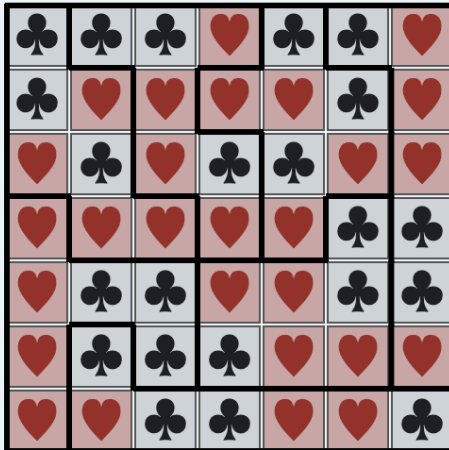
6. Compute the efficiency gap of this district plan, where efficiency gap eg is the difference in the parties' wasted votes divided by the total number of votes.
 - All votes for a losing candidate in a district are wasted.
 - To win a district, 4 votes are needed. Any excess votes for the winner are wasted.
 - For example, if $v_D(\spadesuit) = 2$ and $v_D(\heartsuit) = 5$ in district D , $\heartsuit$ won with $w_D(\heartsuit) = 1$ vote and $w_D(\spadesuit) = 2$ votes.
 - Thus,

$$eg = \left| \frac{\sum_D w_D(\spadesuit) - \sum_D w_D(\heartsuit)}{49} \right|.$$

References

- [Gridtopia Game](#)
- [Outlier Analysis](#)

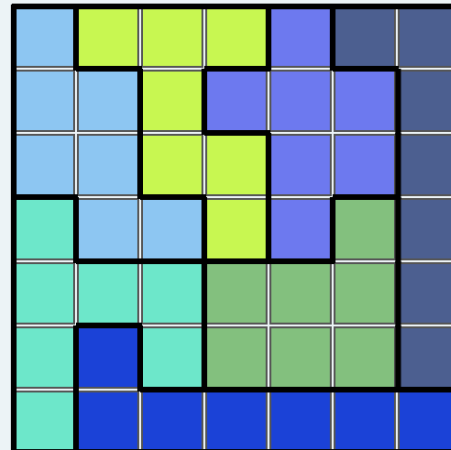
Design a distribution Δ



RANDOMIZE Δ

Click cells to toggle

Design a plan $\mathcal{D}$



RANDOMIZE $\mathcal{D}$

Click cells change their district assignments

Is your plan an outlier for this distribution?

SAMPLE WITH MCMC

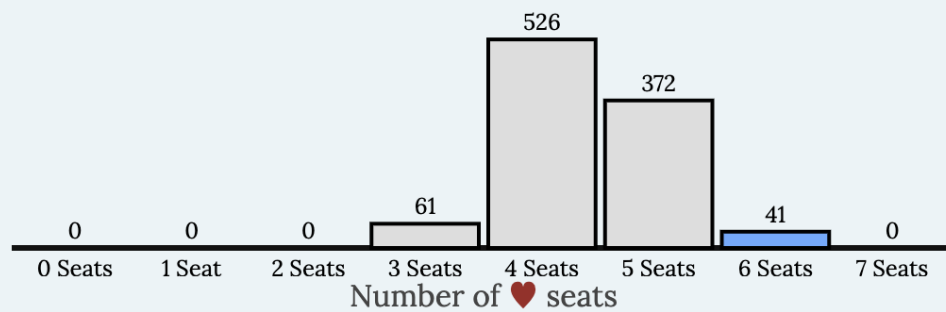


Figure 1: Gridtopia Scenario